

NAME: _____

CLASS: _____



DANEBANK

An Anglican School for Girls

2008

Year 12

Half Yearly Examination

Mathematics

Outcomes Examined: P2,P3,P4,P6,P7,P8,H2, H4, H5, H6,H7,H8,H9

Weighting of Task: 25%

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- All necessary working should be shown in every question otherwise full marks may not be awarded.
- Board-approved calculators may be used
- Start each new question in a new booklet

Total Marks – 84

Attempt All Questions 1 - 7

Questions are of equal value

This paper MUST NOT be removed from the examination room

Question	1	2	3	4	5	6	7
Mark							
Maximum	12	12	12	12	12	12	84

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

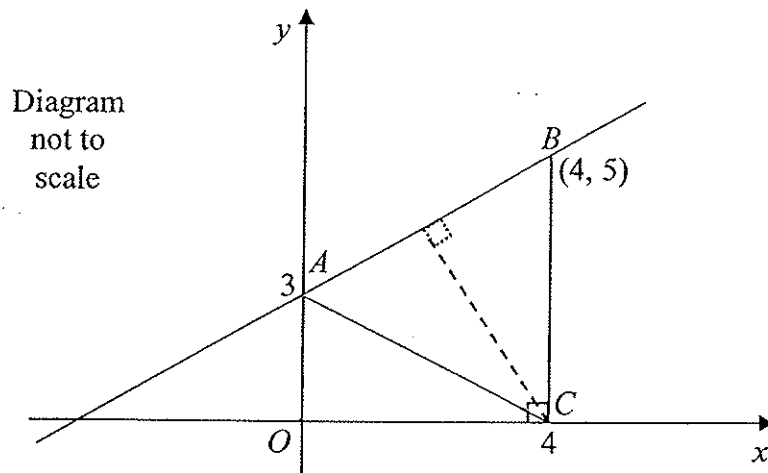
Question 1 (12 marks) Use a SEPARATE writing booklet.	Marks
(a) Evaluate to 2 significant figures – $\frac{\sqrt{4.17^2 - 3.9}}{\pi}$	2
(b) Find the primitive of $x^3 - 2x + 5$	2
(c) Factorise $8a^3 + 1$.	
(d) Prove that $\frac{\sin \theta - \sin^3 \theta}{\cos^3 \theta} = \tan \theta$	2
(e) A parabola satisfies all the following conditions. i. It is concave up; ii. the focal length is one unit iii. the directrix has equation $y = 1$	2
Draw a possible diagram to show this information AND give a possible equation.	
(f) Write $1 + 4 + 7 + \dots + 19$ as an expression in the form $\sum_{n=0}^6 \dots$	2

Question 2 (12 marks) Use a SEPARATE booklet.

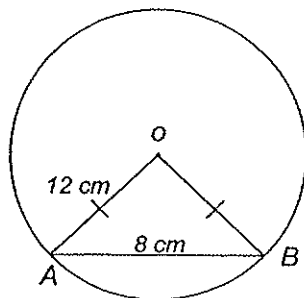
- (a) In the diagram, the line AB cuts the y -axis at the point $A(0, 3)$ and passes through the point $B(4, 5)$. A perpendicular is dropped from B to meet the x -axis at $C(4, 0)$.

Marks

Copy or trace the diagram into your working booklet.



- | | | |
|-------|---|---|
| (i) | Calculate the length of the interval AB . | 1 |
| (ii) | Find the gradient of the line AB . | 1 |
| (iii) | Show that the equation of the line AB is $x - 2y + 6 = 0$ | 1 |
| (iv) | Find the equation of the line which is perpendicular to AB and which passes through C . | 2 |
| (v) | Calculate the perpendicular distance from C to AB . | 2 |
| (vi) | Find the area of the triangle ABC . | 1 |
- (b)



In the diagram $AB = 8\text{ cm}$ and radius of the circle is 12 cm .

- | | | |
|------|---|---|
| (i) | Show that $\angle AOB = 39^\circ$ to the nearest degree. | 2 |
| (ii) | Find the area of the triangle, correct to the nearest cm^2 . | 2 |

Question 3 (12 marks) Use a SEPARATE writing booklet.

Marks

(a) Differentiate with respect to x .

(i) $(x^3 - 1)^5$ 2

(ii) $\frac{4x^2}{x+3}$ 2

(b) (i) Find $\int \frac{2x + x^4}{x^3} dx$. 3

(ii) $\int_1^a (2x + 3) dx = 0$, find value(s) of a 3

(c) Find the values of k for which the quadratic equation $kx^2 - 8k + k = 0$ has real roots. 2

Question 4 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Explain (DO NOT FIND) how you can show that $y = 5x - 16$ is a tangent to the parabola $y = x^2 - 3x$.

1

- (b) Allia puts \$100 into a fund at the end of each month, which earns 7.2% interest per annum compounded monthly.

3

Calculate the amount she will receive if she takes her investment out at the end of 5 years.

- (c) The gradient of a curve is given by $\frac{dy}{dx} = 3x^2 - 6x - 9$. and the curve passes through the point (1,-2).

(i) Show the equation of the curve is $y = x^3 - 3x^2 - 9x + 9$.

2

(ii) Find the co-ordinates of the stationary points and determine their nature.

2

(iii) Find the co-ordinates of the point of inflexion

1

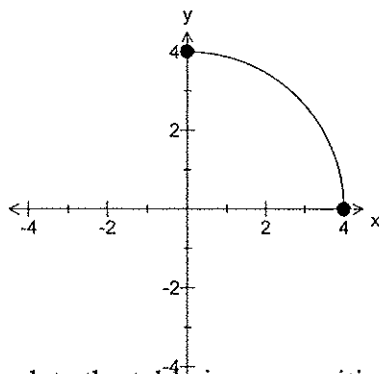
(iv) Sketch the curve of $y = x^3 - 3x^2 - 9x + 9$.
for the domain $-2 \leq x \leq 4$.

3

Question 5 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) The graph below represents $y = \sqrt{16 - x^2}$, $0 \leq x \leq 4$



- (i) Copy and complete the table in your writing booklet. Leave answers to three decimal places.

1

x	0	1	2	3	4
y	4		3.464		

- (ii) Use Simpson's rule with five function values to estimate $\int_0^4 \sqrt{16 - x^2} dx$.

2

- (iii) Find the exact value of $\int_0^4 \sqrt{16 - x^2} dx$ in terms of π .

1

- (iv) Hence, use the answers to parts (ii) and (iii) to approximate the value of π to 2 decimal places.

1

- (b) (i) What is the condition for a geometric series to have a limiting sum?

1

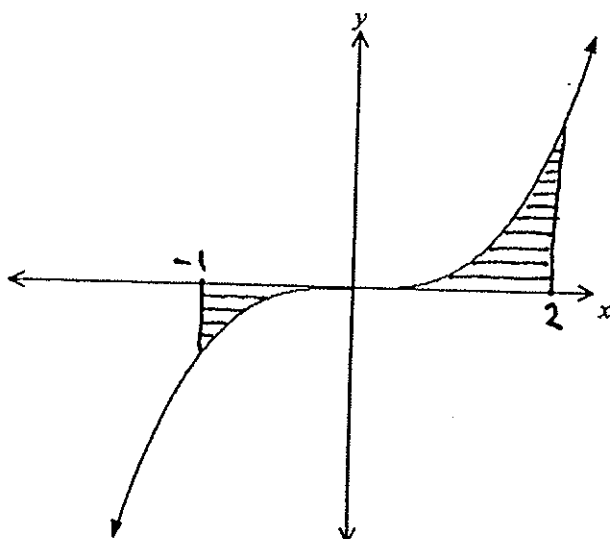
- (ii) The limiting sum of a geometric series is 20.
What might the series be? (Give two possible answers).

2

Question 5 continued over the page

(c) Max was given this question to answer.

“Find the area bounded by the curve $y = x^3$ and the x -axis between $x = -1$ and $x = 2$.



This was his solution:

$$\int_{-1}^2 x^3 dx = \left[\frac{x^4}{4} \right]_{-1}^2 \quad \text{line 1}$$

$$= \left[\frac{2^4}{4} \right] - \left[\frac{(-1)^4}{4} \right] \quad \text{line 2}$$

$$= 63 \frac{3}{4} \text{ units}^2 \quad \text{line 3}$$

Explain fully any errors made and find the actual area showing all working

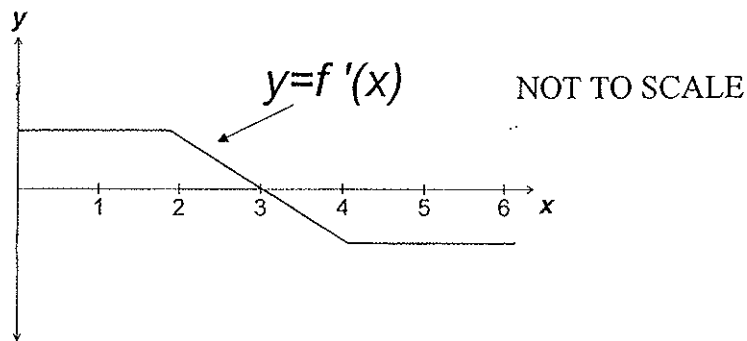
Question 6 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Find the volume of the solid formed when the area bounded by the parabola $y = 4 - x^2$ and the x -axis is rotated about the y -axis.

3

(b)



2

The above diagram shows a sketch of the gradient function of $y = f(x)$. In your writing booklet, draw a sketch of the function $y = f(x)$ given that $f(0) = 0$.

- (c) The eight and fourteenth term of an arithmetic series are -25 and -49 .

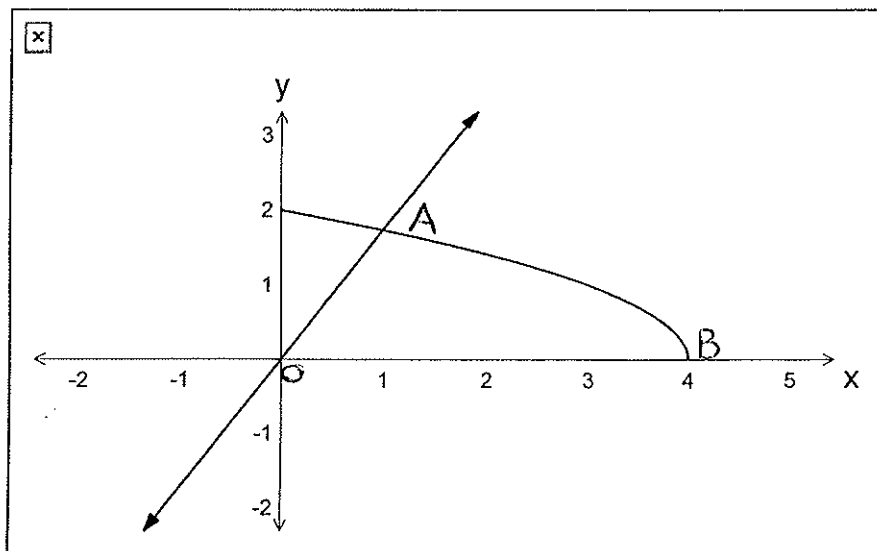
- (i) Find the 1st term and the common difference.

2

- (ii) Find the sum of the first fourteen terms.

1

- (d) The sketch below represents the curve $y = \sqrt{4-x}$ and the line $y = \sqrt{3}x$.



- (i) Show that the point of intersection $A = (1, \sqrt{3})$.

1

- (ii) Hence, find the exact area bounded by OAB and the x axis.

3

Question 7 (12 marks) Use a SEPARATE writing booklet.

Marks

(a) Solve $\sin^2 x = \frac{1}{4}$ for $0^\circ \leq x \leq 360^\circ$.

2

- (b) Statistics have shown that the population of a particular town was decreasing in such a way that at the end of each year the population could be determined as follows.

“10% of the population moved out at the end of the year and 500 people moved in during the year.”

At the beginning of 2000, before the 10% moved out, the population of the town 10 000.

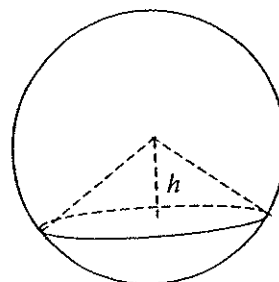
- (i) Show that at the end of 2002 the population of the town was:
 $10\,000(0.9)^3 + 500(1 + 0.9 + 0.9^2)$.

2

- (ii) This trend continued indefinitely. Find the population of the town at the end of the year 2019 (i.e. at the end of the 20th year).

2

- (c) The diagram shows a cone in a sphere.
The vertex of the cone is at the centre of the sphere.
The radius of the sphere is 12cm.



- (i) Show that the volume V of the cone is

2

$$V = \frac{\pi}{3}(144h - h^3)$$

- (ii) Find the maximum volume of the cone in exact form.

4

END OF PAPER

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Half Yearly Solutions 2008 Mathematics

1 (a) $\frac{\sqrt{4 \cdot 17^2 - 3 \cdot 9}}{\pi} = 1.16906429$
 $= 1.2$

Marking Guide:

1 for calculation

2 1 for significant fig

(b) $\int x^3 - 2x + 5 \, dx = \frac{x^4}{4} - x^2 + 5x + C$

2 $\frac{1}{2}$ for each part

(c) $8a^3 + 1 = (2a + 1)(4a^2 - 2a + 1)$

2 1 for progress

(d) To prove $\frac{\sin \theta (1 - \sin^2 \theta)}{\cos^3 \theta} = \tan \theta$

LHS = $\frac{\sin \theta (1 - \sin^2 \theta)}{\cos^3 \theta}$

$= \sin \theta \times \cos^2 \theta$

$\cos^3 \theta$

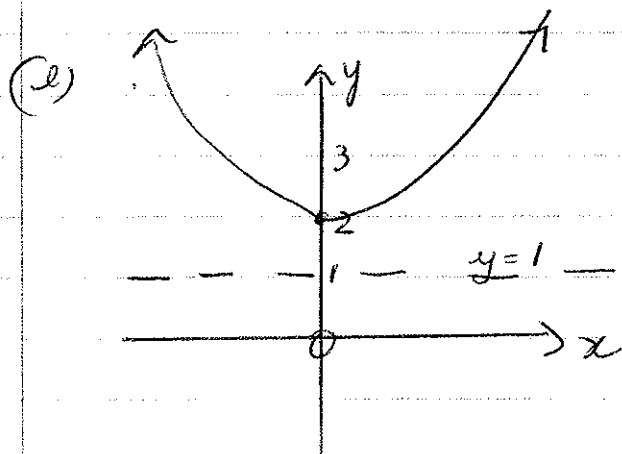
$= \frac{\sin \theta}{\cos \theta}$

$= \tan \theta$

$= \text{RHS}$

1 mark

1 mark



$\frac{1}{2}$ concave up

$\frac{1}{2}$ directrix

$\frac{1}{2}$ equation

$\frac{1}{2}$ vertex

(f) $1 + 4 + 7 + \dots + 19$

$a = 1, d = 3, T_n = a + (n-1)d$ $1 + 6 \times 3$

$T_n = 3n + 1$

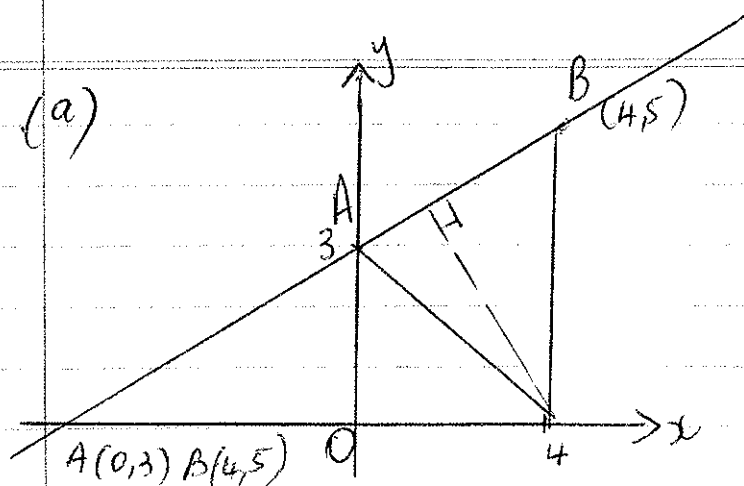
$\sum_{n=0}^6 3n + 1$

1 mark for making

progress

2 answer

2 (a)



Marking guide

$$(i) d_{AB} = \sqrt{(5-3)^2 + (4-0)^2}$$

$$= \sqrt{20}$$

$$d = 2\sqrt{5} \text{ units}$$

$$(ii) m_{AB} = \frac{5-3}{4-0}$$

$$m = \frac{1}{2}$$

$$(iii) \text{Eqn } AB: y-5 = \frac{1}{2}(x-4)$$

$$2y-10 = x-4$$

$$x-2y+6=0$$

$$y-3 = \frac{1}{2}(x-0)$$

$$2y-6 = x$$

$$x-2y+6=0$$

$$(iv) \text{Eqn } \perp: y-0 = -2(x-4)$$

$$y = -2x + 8$$

$$2x + y - 8 = 0$$

$$(v) d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$= \frac{|1 \times 4 - 2 \times 0 + 6|}{\sqrt{1^2 + (-2)^2}}$$

$$= \frac{|4+6|}{\sqrt{5}}$$

$$= \frac{10}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}}$$

$$= 2\sqrt{5} \text{ units, or } 4.47 \text{ (2dp)}$$

$$\frac{1}{2} \times (i) \times (v)$$

$$(vi) \text{Area} = \frac{1}{2} \times 2\sqrt{5} \times 2\sqrt{5}$$

$$= 10 \text{ units}^2$$

1 mark gradient
2 correct eqn

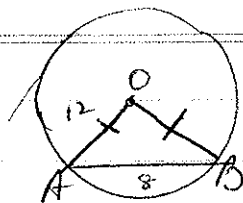
1 mark for wrong
line/wrong point

2 answer

$\frac{1}{2}$ this stage

1 correct answer

(b)



Marking Guidelines

$$(i) \cos \angle AOB = \frac{12^2 + 12^2 - 8^2}{2 \times 12 \times 12}$$

$$= \frac{224}{288}$$

$$\angle AOB = \cos^{-1} \left(\frac{224}{288} \right)$$

$$= 38^\circ \text{ ~~966~~ 24}$$

$$= 39^\circ \text{ to nearest degree}$$

1/2

2 marks

2 marks

$$(ii) \text{ Area} = \frac{1}{2} ab \sin C$$

$$= \frac{1}{2} \times 12 \times 12 \times \sin 39^\circ$$

$$= 45 \text{ cm}^2 \text{ (nearest)}$$

-1 if cos

2 { 1
1

3 (a)

$$\begin{aligned} \textcircled{1} \frac{d}{dx} (x^3 - 1)^5 &= 5(x^3 - 1)^4 \times 3x^2 \\ &= 15x^2(x^3 - 1)^4 \end{aligned}$$

$$\begin{aligned} \textcircled{11} \frac{d}{dx} \frac{4x^2}{x+3} &= \frac{(x+3)8x - (4x^2)1}{(x+3)^2} \\ &= \frac{8x^2 + 24x - 4x^2}{(x+3)^2} \\ &= \frac{4x^2 + 24x}{(x+3)^2} \end{aligned}$$

$$u = 4x^2$$

$$u' = 8x$$

$$v = x+3$$

$$v' = 1$$

Marking Guide

1 mark each part

1 mark for the
1/2 if rule correctly

2 simplify

$$\begin{aligned} \textcircled{b} \textcircled{1} \int \frac{2x + x^4}{x^3} dx &= \int 2x^{-2} + x dx \\ &= -\frac{2}{x} + \frac{x^2}{2} + C \end{aligned}$$

1 mark rewrite
1 correct answer
1 y + c only

$$\begin{aligned} \textcircled{11} \int_1^a (2x+3) dx &= 0 \\ \therefore [x^2 + 3x]_1^a &= 0 \\ (a^2 + 3a) - (1^2 + 3 \times 1) &= 0 \end{aligned}$$

1 mark integrate
1 mark substitute
1 mark solve

$$a^2 + 3a - 4 = 0$$

$$(a+4)(a-1) = 0$$

$$\therefore a = -4, 1$$

$$\begin{aligned} \textcircled{c} \quad kx^2 - 8x + k &= 0 \\ \Delta &= b^2 - 4ac \\ &= (-8)^2 - 4(k)(k) \\ &= 64 - 4k^2 \end{aligned}$$

For real roots $\Delta \geq 0$

$$\therefore 64 - 4k^2 \geq 0$$

$$16 - k^2 \geq 0$$

$$(4-k)(4+k) \geq 0$$

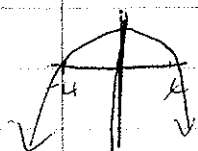
$$\therefore -4 \leq k \leq 4$$

$$\frac{1}{2} \Delta \geq 0$$

$$\frac{1}{2} \Delta = 64 - 4k^2$$

-1/2 if only > not

-1/2 final inequality incorrect



4(a) $y = x^2 - 3x$ and $y = 5x - 16$ have one point of intersection by solving them simultaneously, then the line $y = 5x - 16$ is a tangent to $y = x^2 - 3x$.

Marking Guideline

for reasonable explanation

(b) $P = \$100$ $r = 7.2\% \text{ p.a.}$ $n = 5 \times 12$
 $= 0.6\% \text{ pm}$ $n = 60$
 $= 0.006 \text{ pm}$

-1 if wrong interest rate and n

$A_1 =$ first \$100 invested

$$= \$100 \times 1.006^{60}$$

$$A_2 = \$100 \times 1.006^{59}$$

$$A_3 = \$100 \times 1.006^{58}$$

$$A_{60} = \$100 \times 1.006^1$$

-1 if in wrong order

$$\begin{aligned} \therefore \text{Total} &= \$100 \times 1.006^{60} + \$100 \times 1.006^{59} + \dots + \$100 \times 1.006^1 \\ &= \$100 \times 1.006 (1.006^{59} + 1.006^{58} + 1.006^{57} + \dots + 1) \\ &= \$100 \times 1.006 (\text{sum of geometric series with } a=1, r=1.006, n=60) \\ &= \$100 \times 1.006 \left[\frac{a(r^n - 1)}{r - 1} \right] \end{aligned}$$

1 mark

$$= \$100 \times 1.006 \left[\frac{1(1.006^{60} - 1)}{1.006 - 1} \right]$$

1 mark

$$= \$7239.65$$

$$(i) \frac{dy}{dx} = 3x^2 - 6x - 9$$

$$(1) y = \frac{3x^3}{3} - \frac{6x^2}{2} - 9x + C$$

$$y = x^3 - 3x^2 - 9x + C$$

$$-2 = 1^3 - 3(1)^2 - 9(1) + C$$

$$C = 9$$

$$\therefore y = x^3 - 3x^2 - 9x + 9$$

$$(ii) \frac{dy}{dx} = 3x^2 - 6x - 9$$

$$\frac{d^2y}{dx^2} = 6x - 6$$

Stationary points when $\frac{dy}{dx} = 0$

$$\therefore 3x^2 - 6x - 9 = 0$$

$$x^2 - 2x - 3 = 0$$

$$(x-3)(x+1) = 0$$

$$x = 3, -1$$

$$\text{When } x = 3, y = 3^3 - 3(3)^2 - 9(3) + 9 = -18$$

$$\text{and } \frac{d^2y}{dx^2} = 6(3) - 6 = 12$$

$$> 0$$

$\therefore$ Minimum turning point at $(3, -18)$

$$\text{When } x = -1, y = (-1)^3 - 3(-1)^2 - 9(-1) + 9 = 14$$

$$\text{and } \frac{d^2y}{dx^2} = 6(-1) - 6 = -12$$

$$< 0$$

$\therefore$ Maximum turning point at $(-1, 14)$.

1 for integration

2 { 1 for constant

1 points

1 nature

Marking guide

(iii) Pt of inflexion when $\frac{d^2y}{dx^2} = 0$

$$\therefore 6x - 6 = 0$$

$$x = 1, y = 1^3 - 3(1)^2 - 9(1) + 9 = -2$$

Testing concavity

x	0	1	2
$\frac{d^2y}{dx^2}$	< 0	0	> 0

$\therefore$ change in concavity
 $\therefore (1, -2)$ is a pt of inflexion

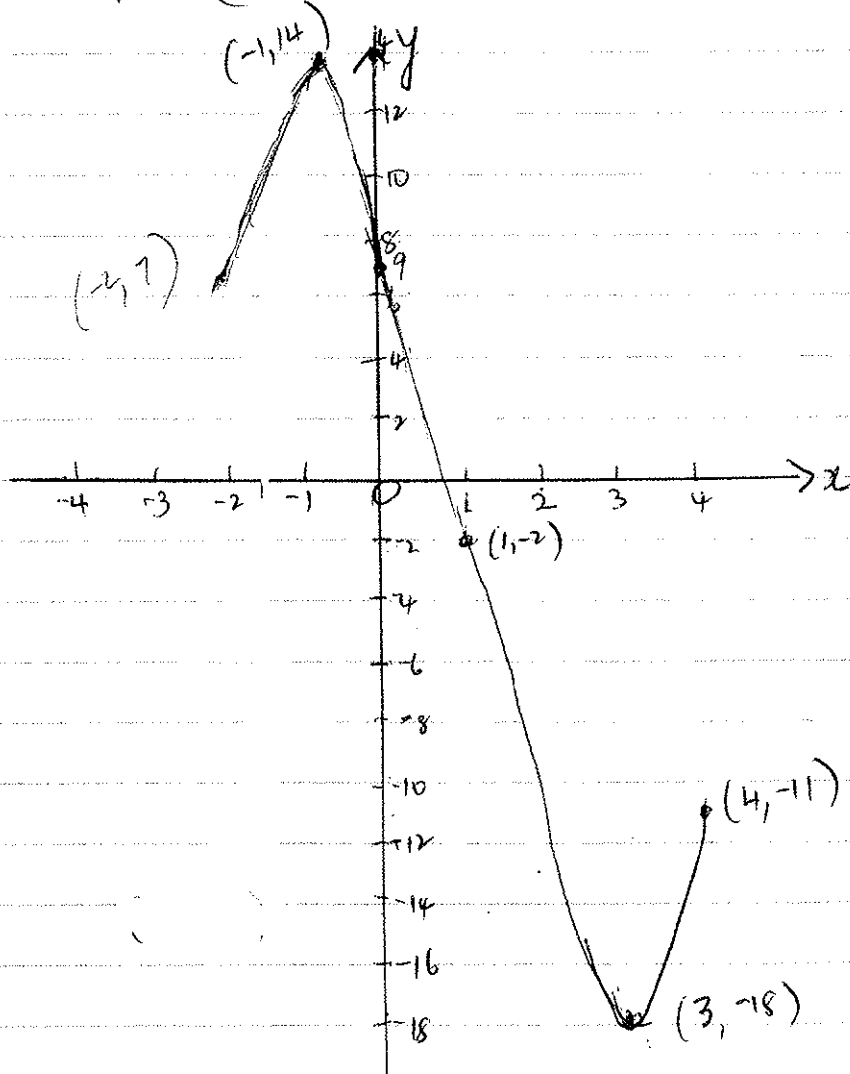
$\frac{1}{2}$ point

$\frac{1}{2}$ test

(iv) End points: When $x = -2$, $y = (-2)^3 - 3(-2)^2 - 9(-2) + 9$
 $= 7$ $\therefore (-2, 7)$

When $x = 4$, $y = 4^3 - 3(4)^2 - 9(4) + 9$
 $= -11$ $\therefore (4, -11)$

y intercept $(0, 9)$



$\frac{1}{2}$ for each end point, turning point, inflexion point & intercept

Marking Guide

5. (a)(i) $y = \sqrt{16-x^2}$

x	0	1	2	3	4
y	4	3.873	3.464	2.646	0

$\frac{1}{2}$ for any error
1 correct answer

(ii) $\int_0^4 \sqrt{16-x^2} dx = \frac{1}{3} [4 + 0 + 4(3.873 + 2.646) + 2(3.464)]$
 $= \frac{1}{3} (4 + 26.076 + 6.928) = 37.004$
 $= 12.3346$
 $= 12.335$ (3dp)

1 mark

2 answer

(iii) $\int_0^4 \sqrt{16-x^2} dx = \frac{1}{4} \pi (4)^2$
 $= 4\pi$

1

(iv) $\therefore 12.337 = 4\pi$
 $\therefore \pi = \frac{12.335}{4}$
 $= 3.08375$
 ≈ 3.08 (to 2dp)

1

(b) (i) $-1 < r < 1$

1

(ii) $S_{\infty} = \frac{a}{1-r}$

$20 = \frac{a}{1-r}$

$20(1-r) = a$

1 mark

If $r = \frac{1}{2}$, $20(1 - \frac{1}{2}) = a$
 $\therefore a = 10$

$\therefore$ Series could be $10, 5, 2\frac{1}{2}, \dots$

If $r = -\frac{1}{2}$, $20(1 + \frac{1}{2}) = a$
 $a = 30$

$\therefore$ Series could be $30, -15, 7\frac{1}{2}, \dots$

1 each series

1 mark

(c) Error, is in line 3 with the calculation $3\frac{3}{4}$ units.

$$\begin{aligned}\text{Correct working Area} &= \left| \int_{-1}^0 x^3 dx \right| + \int_0^2 x^3 dx \\ &= \left| \left[\frac{x^4}{4} \right]_{-1}^0 \right| + \left[\frac{x^4}{4} \right]_0^2 \\ &= \left(\frac{0}{4} - \left(\frac{-1}{4} \right)^4 \right) + \left(\frac{2^4}{4} - 0 \right) \\ &= 4\frac{1}{4} \text{ units}^2\end{aligned}$$

• You must take absolute value of area under the curve.

Marking Guideline

1

1

1

Marking Guideline

6. (a) $V = \pi \int_0^4 x^2 dy$ $y = 4 - x^2$
 $x^2 = 4 - y$



$$= \pi \int_0^4 (4 - y) dy$$

$$= \pi \left[4y - \frac{y^2}{2} \right]_0^4$$

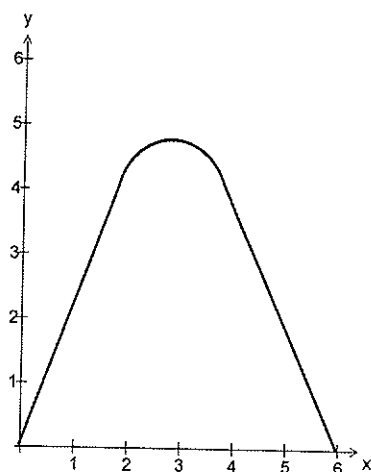
$$= \pi [(16 - 8) - (0 - 0)]$$

$$= 8\pi$$

3.

1 for $\int_0^4$
 1 for $x^2 = 4 - y$
 $\frac{1}{2}$ correct integration
 $\frac{1}{2}$ correct answer

(b)



2.

$\frac{1}{2}$ for identifying
 2 straight lines
 with correct
 gradients.
 $\frac{1}{2}$ identifying parabola
 $\frac{1}{2}$ with max
 turning point

(c)

(1) $T_8: a + 7d = -25$ ①

$T_{14}: a + 13d = -49$ ②

② - ① $6d = -24$

$d = -4$

$\therefore a + 7(-4) = -25$

$a = 3$

2.

1 for 2 correct
 equations
 1 mark correct
 answer.

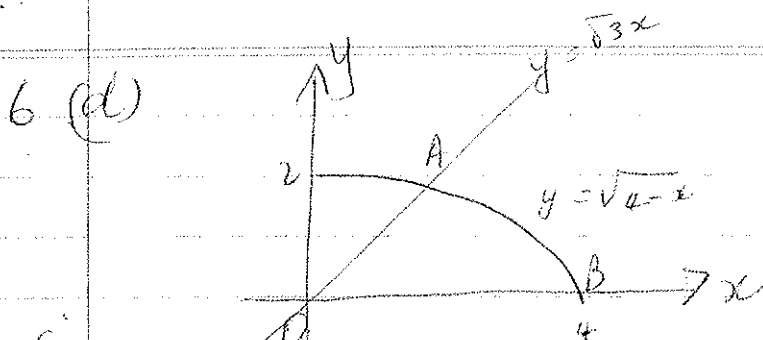
(ii) $S_n = \frac{n}{2} [2a + (n-1)d]$ $a = 3$

$= \frac{14}{2} [2 \times 3 + 13 \times -4]$ $d = -4$

$n = 14$

$= -322$

1 for a, d, n
 2 answer



(i)

$y = \sqrt{3}x$ & $y = \sqrt{4-x}$ solve simultaneously
 $\therefore \sqrt{4-x} = \sqrt{3}x$ $\leftarrow \frac{1}{2}$

Squaring $4-x = 3x^2$
 $3x^2 + x - 4 = 0$

$(3x+4)(x-1) = 0$

$\therefore x = -4/3, 1$

When $x=1$, $y = \sqrt{3}(1)$

$\therefore A$ is $(1, \sqrt{3})$

1 mark for correct working

(ii) Area = $\int_0^1 \sqrt{3}x dx + \int_1^4 \sqrt{4-x} dx$ $\frac{(4-x)^{3/2}}{3/2}$
 $= \left[\frac{\sqrt{3}x^2}{2} \right]_0^1 + \left[\frac{(4-x)^{3/2}}{-3/2} \right]_1^4$
 $= \left[\frac{\sqrt{3}}{2} (1) - 0 \right] + \frac{2}{3} \left[(4-4)^{3/2} - (4-1)^{3/2} \right]$
 $= \frac{\sqrt{3}}{2} + \frac{2}{3} (0 - 3^{3/2})$
 $= \frac{\sqrt{3}}{2} + \frac{2}{3} 3\sqrt{3}$
 $= \frac{\sqrt{3}}{2} + 2\sqrt{3} \text{ units}^2$
 $= \frac{5\sqrt{3}}{2} = 4.3 \text{ units}^2$

1 mark

1 correct integration

1 answer

4.3 (really)

$$7(a) \sin^2 x = \frac{1}{4}$$

$$\sin x = \pm \frac{1}{2}$$

$$x = 30^\circ, 150^\circ, 210^\circ, 330^\circ$$

Marking Guide

1. solution's

2. answers

(-1/2 for omissions)

(b) (i) ^{start} 2000 $\rightarrow$ population of 10 000

end 2000 after 1 year population is

$$\text{end 2000 } A_1 = 10\,000 \times 0.9 + 500$$

$$\text{end 2001 } A_2 = A_1 \times 0.9 + 500$$

$$= (10\,000 \times 0.9 + 500) \times 0.9 + 500$$

$$= 10\,000 \times 0.9^2 + 500(1 + 0.9)$$

$$\text{end 2002 } A_3 = A_2 \times 0.9 + 500$$

$$= [10\,000 \times 0.9^2 + 500(1 + 0.9)] \times 0.9 + 500$$

$$= 10\,000 \times 0.9^3 + 500(1 + 0.9 + 0.9^2)$$

$$A_{20} = 10\,000 \times 0.9^{20} + 500(1 + 0.9 + 0.9^2 + \dots + 0.9^{19})$$

$$= 10\,000 \times 0.9^{20} + 500(\text{sum of geometric series})$$

$$= 10\,000 \times 0.9^{20} + 500 \left[\frac{a(1-r^n)}{1-r} \right] \quad \begin{matrix} a=1, n=20 \\ r=0.9 \end{matrix}$$

$$= 10\,000 \times 0.9^{20} + 500 \left[\frac{1(1-0.9^{20})}{1-0.9} \right]$$

$$= 1215.766 + 500 \times 5.17949$$

$$= 5607.883273$$

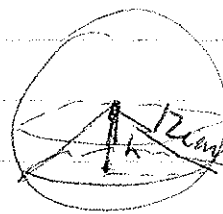
end of 2019 = 5608 people

$$(c) (i) r = \sqrt{144 - h^2}$$

$$V = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \pi (\sqrt{144 - h^2})^2 \cdot h$$

$$= \frac{\pi}{3} (144h - h^3)$$



$$(ii) \frac{dV}{dh} = \frac{\pi}{3} (144 - 3h^2)$$

$$\frac{d^2V}{dh^2} = \pi - 9h$$

$$= -3\pi h$$

$$< 0$$

$\therefore$ Max for all h since $h > 0$

Marking Guide

Maximum volume when $\frac{dV}{dh} = 0$

$$\frac{\pi}{3} (144 - 3h^2) = 0$$

$$3h^2 = 144$$

$$h^2 = 48$$

$$h = \pm\sqrt{48}$$

Since $h > 0$ for length

$$h = 4\sqrt{3}$$

$$\therefore \text{Volume} = \frac{\pi}{3} (144 \times 4\sqrt{3} - (4\sqrt{3})^3)$$

$$= \frac{\pi}{3} (576\sqrt{3} - 192\sqrt{3})$$

$$= 128\sqrt{3} \pi \text{ cm}^3$$